Chest X-Ray Pneumonia Detection Using Convolutional Neural Networks: A Comparative and Explainable Study of a Custom CNN, VGG16, and ResNet50

Mani Teja Padala
Student ID: 69767331
Department of Business, University of Europe for Applied Sciences,
Konrad-Zuse-Ring 11, Potsdam, 14469, Brandenburg, Germany

Abstract

Pneumonia is a leading cause of death globally, particularly for children under 5 years of age and is diagnosed on chest X-rays requiring access to highly scarce expert radiologists. This work is aimed at automatic binary classification of paediatric chest X-rays into Normal and Pneumonia using CNN. There is a lack of consistent and reasonable comparison of custom networks versus pre-trained networks, both in the way classes are handled and in terms of efficiency, when class-imbalance is considered. Three models were built and tested on the Kermany Chest X-ray dataset comprising 5,856 images: a custom CNN model, a VGG16 transfer-learning model, and a ResNet50 transfer-learning model, all of which were augmented and balanced for class weighting. An independent test set of 624 images was used to evaluate the models in terms of accuracy, precision, sensitivity, specificity, F1-score, AUC-ROC, and efficiency, the results of which were interpreted using Grad-CAM. VGG16 was the most successful model, with a 92.5% accuracy, 98.2% sensitivity, and an AUC-ROC of 0.974, making an error of just 7 out of 390 pneumonia cases. Although the smallest and fastest model, the custom CNN performed the poorest with 62.2% accuracy and 41.5% sensitivity. The results indicate that transfer learning, especially VGG16, is much more robust and clinically safer for pneumonia screening than a shallow custom network, on which the key criterion is sensitivity. The study provides a reproducible, explainable and efficiency-aware comparison and suggests VGG16 as a model which can be used as a screening-support model with human supervision.

Keywords: The proposed system is based on Convolutional Neural Networks (CNN), Transfer Learning, and Pneumonia Detection.The proposed system uses Convolutional Neural Networks (CNN) and Transfer Learning for the detection of pneumonia in a chest X-ray image.

1 Introduction

The problem context, as well as its application value, will be discussed.Problem's context and its practical value will be discussed.

Pneumonia is an acute respiratory infection in which air sacs in the lungs become inflamed and is still one of the world's most common causes of death. It is particularly hazardous for children under five years of age, where it constitutes a significant proportion of child mortality around the world every year [1]. Currently, the diagnostic course of action involves a trained radiologist visual inspection of a chest X-ray for the appearance of lung opacification typical of infection [2]. In many low and middle income areas, however, there is a critical problem with the availability of qualified radiologists, which means that diagnosis and treatment is delayed [3]. Manual interpretation also is subjective, labor intensive and subject to inter-observer variability, especially in high volume cases [4]. These delays are clinically significant as pneumonia can rapidly progress and early antibiotics or supportive treatment significantly improve survival [1]. A computer-based system that identifies suspected pneumonia from a chest radiograph would, therefore, help decrease the diagnostic delay and alleviate the burden of the limited number of expert staff members. This system is not designed to replace the clinicians but it is designed to be used as a decision support; high risk cases would be reviewed. Due to the high clinical impact and the lack of specialists, automated chest X-ray analysis is a practically important problem [3].

In this section, we will provide some background on machine learning and computer vision.

One of the most common types of machine learning is image-based machine learning, where a computer learns predictive patterns directly from the pixel data instead of through hand-crafted rules [5]. For medical image classification, convolutional neural networks (CNNs) are the prevalent solution as they automatically learn hierarchical visual features [6]. Lower layers are good at identifying lower level features, including edges and textures, while higher layers are good at identifying higher level features, such as those relevant to the disease [6]. This automatic feature extraction eliminates the need for manual feature engineering, a complex task and fragile for complex medical images [5]. CNNs can also learn the diffuse opacity and consolidation patterns in chest radiographs to differentiate pneumonia from normal lungs [7]. CNNs also make use of spatial structure in the form of local connectivity and weight sharing, which could result in the efficiency of the CNN and robustness to small translations [6]. One of the main difficulties in medical imaging is, however, the scarcity of good quality labeled data, due to the high cost of expert annotation [8]. Few data increases the risk of overfitting, where the model learns the training set rather than generalising patterns [8]. This is why there are methods like data augmentation, class balancing, transfer learning, and others that are at the heart of this study [9].

In the selected problem the transfer learning is performed.

However, deep CNN training via random initialisation usually needs huge amounts of labeled data, which are difficult to obtain in medical fields [9]. To overcome this, transfer learning techniques have been developed to re-use a network trained on a large general purpose dataset like ImageNet [10]. The pre-trained network has already acquired a wide variety of visual features, like edges and textures, as well as shapes, which are useful for new tasks [10]. In practice, the pre-trained convolutional base is shared as a feature extractor and a new classification head is trained for the target task, thus making the model reusable across different tasks [11]. The network can then be adapted to the actual medical images by unfreezing some of the later layers to make the features more specific to the medical images [11]. This is typically more accurate and more quickly converging than training a network from scratch, when data is limited [9]. The architectures of VGG16 and ResNet50 are chosen as transfer-learning backbones in this study because they are widely used in computer vision. The selection of these models was motivated by their solidity, good documentation and their success in medical image applications in the past [7]. Each architecture is described in detail in the Methodology as recommended by the report structure.

The concepts of explainable AI and trustworthiness will be introduced.Explainable AI and trustworthiness will be explained.

Even if the model had a high predictive accuracy, it would not be enough to justify putting the model to use in a clinical setting [12]. A model can be scoring well, but that does not mean it is learning features of the disease, it could be using spurious shortcuts, such as text markers or background artefacts, in order to obtain the highest score [13]. Explainable AI methods are thus required to verify that the predictions are based on clinically relevant parts of the image [12]. Grad-CAM generates class-discriminative heatmaps that provide an indication of the areas most important for a prediction [14]. SHAP offers a theory-backed, complementary attribute of the influence of image regions on a prediction to one class or the other [15]. When combined these techniques enable a developer or clinician to check whether the model is looking at the lung fields or not paying attention to irrelevant areas [13]. In a high-stakes medical environment, where an incorrect or incorrect direction prediction could have serious consequences, this is especially important [12]. However, for the detection of pneumonia, a false negative may lead to the delay of treatment for a truly sick patient and hence the model's reasoning process needs to be trusted [1]. There is therefore a direct link between explainability and the safe and responsible use of automated screening tools [12].

1.1 Research Gap

CNN-based pneumonia detection has been shown to have high accuracy in various studies, but there are still some methodological gaps [7]. Most studies test just one architecture and typically compare a custom CNN with several transfer-learning models in different settings without a direct comparison in the same setting [16]. Many studies report the accuracy of their model on the top level without properly accounting for the large class imbalance found in chest x-ray datasets [17]. In particular, the most clinically important measure of a screening test, sensitivity, is sometimes not reported in favor of accuracy [4]. There are relatively few studies that integrate predictive evaluation and explainability analysis with more than one attribution method [13]. Computational efficiency, such as the model size and inference speed, is also often neglected in spite of its key role in deployment for actual applications [18]. This study addresses these gaps by performing a systematic comparison of a custom CNN with two transfer-learning models on a common experimental protocol, which is designed to be imbalanced-aware. It also combines with Grad-CAM and SHAP explanations and provides efficiency metrics along with predictive performance.

1.2 Problem Statement

The main challenge that we are tackling in this research is automatic binary classification of paediatric chest X-ray images into Normal and Pneumonia. The Normal class is healthy lungs with no signs of infection on the radiograph; and the Pneumonia class is lungs with bacterial or viral pneumonia. Class imbalance is one of the key challenges in the principal modelling as the number of pneumonia cases in the available data set is about 73Another problem is that pneumonia can look like other lung disease conditions and the circumstances are slight and overlapping. This is a high stakes medical domain and false negative (missed pneumonia) is dangerous and needs to be minimised. The problem in this study is therefore to classify automatically the chest X-ray images with a comparative and explainable CNN-based framework. The study explores the possibility of using transfer learning to enhance predictive performance compared to a custom CNN whilst simultaneously keeping the computational efficiency of the CNN and generating meaningful visual explanations.

1.3 Aim and Objectives

This research aims to design and test an explainable CNN based approach for reliable classification of chest X-ray images as Normal or Pneumonia. The following objectives will help to achieve this aim:

- process the public Chest X-ray (Pneumonia) dataset, analyze it, and preprocess it (also address class imbalance);

- train a CNN from scratch and use it as a baseline; - train two transfer learning models: VGG16 and ResNet50; and
- compare the accuracy, precision, sensitivity, specificity, F1-score and AUC-ROC of predictive performance;
- analyse how efficient and how they learn respectively, each model is;
- interpret model predictions with Grad-CAM and SHAP; and
- discuss faults and weaknesses and the feasibility of the system for screening support.

1.4 Research Questions

The following research questions are used to guide this study:

RQ1: Can a custom CNN model trained from scratch, obtain clinically acceptable accuracy ($> 90\%$) of chest X-ray images as Normal or Pneumonia? item **RQ2:** What are the accuracy, sensitivity, specificity and AUC-ROC of transfer learning (VGG16 and ResNet50) vs a custom CNN? What are the effects of data augmentation approaches on the performance of the models trained by an imbalanced medical image dataset?

- **RQ4:** What areas in the chest X-ray images are the most critical for the model's classification, as seen in the Grad-CAM visualization? What is the compromise between the predictive performance and the computational efficiency (model size and inference time) from the three models presented and do the SHAP-based explanations match up with Grad-CAM in marking the influential regions?

1.5 Contributions and Novelty

This research has several contributions regarding the implementation of CNNs to detect pneumonia in a practical and reproducible way. It does not suggest a new architecture, but is based on a careful, fair, and explainable comparison of the two under the same experimental protocol. It is specific contributions as follows:

The end-to-end image-analysis pipeline, from preprocessing through augmentation, class balancing, training, to evaluation, is reproducible; and

- a systematic comparison of the custom CNN with two models for transfer learning (VGG16 and ResNet50) under the same conditions taking imbalance into account; and
- a class-wise reliability analysis with a focus on sensitivity as the clinically relevant measure, instead of accuracy;
- a dual-method explainability analysis with Grad-CAM and SHAP to evaluate if the models focus on clinically relevant areas; and An efficiency analysis that reports the number of parameters, model size and the time it takes to make an inference to evaluate deployment suitability.

1.6 Report Organization

The remainder of this report is organised as follows. Related works on image-based machine learning, CNN and transfer-learning, image preprocessing, evaluation and explainability are reviewed in Section 2 and a comparative literature matrix is presented. The methodology, which encompasses the dataset, preprocessing, model development, training procedure, evaluation framework, and explainability approach, are described in Section 3. The results of the experiments are presented in section 4, following the research questions. The findings, limitations and future directions are discussed and interpreted in Section 5, which also compares them with previous work. The report ends with an overview of the key findings and implications in section 6.

2 Literature Review

The papers are organized around the research theme and not by paper. It introduces image-based machine learning in the medical field, detects pneumonia by using CNN, transfer-learning architectures, introduces the concept of data preprocessing and augmentation, model evaluation and reliability, and explainable AI, and then consolidates the gap to be filled by this study.

Subsection: Image Based Machine Learning in the Medical Domain

In the field of medical diagnosis, image-based machine learning has gradually superseded the traditional, hand-designed image-processing pipelines [8]. The first computer-aided detection systems were based on features that were manually designed, e.g., edges, textures, intensity histograms, and proved to be brittle and difficult to generalize [5]. The era of automatic feature learning directly from pixel data was ushered in by the introduction of deep convolutional neural networks [6]. This transition made it possible for models to learn diffuse opacity patterns typical of pneumonia, without having to be explicitly programmed with rules for the condition [7]. In the field of medical imaging, surveys of deep learning show a significant and consistent improvement in the accuracy of many tasks over classical methods [8]. Concurrently, there are several challenges that have been identified in these surveys, such as small amounts of labelled data, class imbalance, and the requirement for interpretability [7]. This study operates in this ”gray zone” context, using deep CNNs for a binary pneumonia-detection task, and explicitly tackling imbalance and explainability.

In this section, we explore the CNN-based approaches for detecting pneumonia.Next, we will discuss CNN-based methods for pneumonia detection.

The Kermany paediatric chest X-ray dataset has been used to develop a large body of literature that has employed CNNs to the collection and related datasets [1]. Based on the work done by Rajpurkar et al. by CheXNet, they proved that a deep DenseNet can perform at par or even best than the average radiologist in detecting pneumonia, making deep architectures a viable diagnostic aid [19]. The effectiveness of custom CNNs was compared against pre-trained CNNs and that pre-trained CNNs generally demonstrated higher accuracy on the same data set as shown in subsequent studies [20]. The reported accuracies span a wide range, depending on the split of the data sets, pre-processing, and evaluation protocol, making it difficult to make direct numerical comparisons between studies [7]. There are several works that have pointed out the possible misleading accuracy in headlines on this dataset due to the large class imbalance that exists in the data [17]. A frequent methodological drawback is the use of accuracy, neglecting to pay attention to the most clinically relevant metric, sensitivity [16]. This review does not provide a detailed description of each study but rather a comparative description in a literature matrix (Table 2.1).

2.1 Transfer Learning Architectures

In the case of small datasets, one of the most common approaches for pneumonia detection is transfer learning using ImageNet-pretrained backbones [9]. The uniform small convolutional filter stack used by VGG16 is easy to implement, stable and efficient for medical fine-tuning, which is the major reason for its usage [21]. ResNet50 proposes a residual network with residual connections, which can be trained to be extremely deep without having vanishing gradients, enabling it to have a much greater representation capacity [22]. Comparative studies give mixed results between these backbones, and in some cases, VGG16 can perform competitively or best on fewer layers [16, 23]. Other architectures are also tested, including DenseNet121, InceptionV3, Xception and EfficientNet, which can sometimes achieve better performance than VGG and ResNet on specific datasets (see [18, 24]). The performance of these backbones is relative and depends on the data used, therefore these two models are tested under same conditions and same data. In line with the report guidelines, only the architectures actually used in this study (VGG16 and ResNet50) are examined in methodological detail.

Data Preprocessing Data Augmentation

The usual steps performed in the preprocessing phase of chest X-ray classification involve resizing to a fixed input size, converting to the RGB color space, and pixel normalisation [7]. To enhance generalisation, and to partially alleviate class imbalance, data augmentation techniques such as rotation, shifting, zoom, flipping and adjustment of brightness are often applied to the images [9]. Some transformations are not suitable to be applied if the orientation or laterality has diagnostic significance, and augmentation should be carefully selected so as to avoid inappropriate use [7]. In the case of pneumonia detection, moderate geometric and photometric augmentations are generally believed to be safe, as they represent realistic variation in acquisition.In the case of pneumonia detection, it is generally believed that moderate geometric and photometric augmentations are safe as they represent realistic variation in acquisition [16]. Class-imbalance handling has been specifically addressed by class weighting, oversampling, targeted augmentation and weighted and focal loss [17]. However, Chamseddine et al. explicitly used SMOTE in conjunction with a weighted loss to deal with imbalance in chest X-ray classification and they found that minority class performance improves [17]. The current study uses the combination of augmentation and balanced class weighting, as it is widely used and one that is truly effective for this data set.

The evaluation of the models and their reliability is described as follows:

Pneumonia models are assessed using parameters as accuracy, precision, recall or sensitivity, specificity, F1-score and the area under the ROC curve [7]. There are several reviews of evaluation practice that alert to the unreliability of accuracy in class imbalance, and explicitly call for reporting of sensitivity and AUC.A couple of reviews of evaluation practice alert to the unreliability of accuracy in class imbalance, and call for the explicit reporting of sensitivity and of AUC [25]. Some studies also mention the PR-AUC and the best-F1 operating point in order to provide better insight of performance for imbalanced conditions [23]. There are several works that use a single train/validation/test split without running multiple times or conducting statistical tests [7]. Patient-wise splitting is indicated as important so as to avoid leakage of data between training and test sets in Kermany dataset (Data provided at image level [23]). This study presents all the classification measures explicitly with a focus on sensitivity as well as efficiency measures, but does not claim to be a single-split study.

The application of Explainable AI in Image Analysis is discussed.The use of Explainable AI in the field of image analysis is explained.

In the realm of clinical deep-learning systems, explainable AI has become an expected norm, as black-box predictions are hard to believe [12]. The most popular one is Grad-CAM, which generates class-discriminative heat maps showing regions that affect a prediction [14]. It has been widely used in pneumonia detection to verify that models focus on lung areas and not on artefacts of the background [26]. However, researchers warn that even if visually pleasing heatmaps are generated, they do not necessarily indicate correct reasoning, as the heatmaps obtained by saliency methods can be unstable and sensitive to preprocessing [13]. The complementary game-theoretic explanation of individual region contributions to a prediction is provided by SHAP, which is more expensive to compute than Grad-CAM [15]. Comparative explainability studies have revealed that different backbones can give different attention, some of which may be to non-pathological regions [24]. Therefore, in this study, Grad-CAM is used as the main explainability technique to explore the attention paid to clinically relevant lung regions.

A Comparative Literature Review matrix is provided.

Table 2.1 compares a set of similar peer-reviewed studies on the datasets, models, how to deal with imbalance, multi-model comparison, explainability, robustness, external validation, and evaluation metrics. There are a number of common themes in the literature reviewed. Almost universally, transfer learning is applied and sometimes multi-model comparison is applied, while explicit class-imbalance treatment is reported inconsistently. Grad-CAM is quite common while other complementary attribution methods, such as SHAP, are relatively scarce. Robustness testing and external validation to other data sets are rarely reported, thus a methodological gap. The last row of the table places the present study in the context of a fair comparison between custom vs. pretrained models, explicit handling of imbalance, explainable via Grad-CAM, and efficiency analysis all in a single reproducible pipeline.

The synthesis of literature and the identified gap will be presented.

When synthesising the reviewed studies, a uniformity of strengths and weaknesses is revealed. Transfer learning has consistently been demonstrated to yield superior performance to training from scratch on the Kermany dataset, and comparing multiple models has become a common practice. In particular, due to the focus on interpretability in recent models, Grad-CAM is often applied to explain the model's outputs visually. Some gaps are, however, common in studies, and the focus of this work is directly addressing these gaps. The dataset is highly skewed (73:27), but the treatment is not applied explicitly or rigorously in the classes. Limited and inconsistent visual evidence of models relying on clinically relevant regions is reported, and there is often only one attribution method that is reported. Being able to deploy an efficient computational model, including model size and inference time, is often neglected but is crucial for some deployments. This study aims to fill these gaps by providing a fair comparison between the custom model and the pretrained model, explicit handling of imbalances, explainability using Grad-CAM, and efficiency analysis, which motivates the methodology presented in the next section.

3 Methodology

3.1 Research Design

The type of research design used in this study is experimental research design and quantitative research design. The image task is a binary classification of chest X-ray images to determine whether the image is containing pneumonia or not containing pneumonia. The work is explicitly comparative, where one custom CNN is compared to two transfer-learning models under identical conditions. Three models are tested: A custom trained CNN from scratch, a transfer learning CNN VGG16, and a transfer CNN Resnet50. The Grad-CAM Explainable AI method is used to measure the model interpretability. The evaluation is conducted on a separate test set with a wide range of classification and efficiency metrics. The overall workflow, summarizing the entire pipeline

Comparative literature review matrix: Summarising the datasets used, models, handling of imbalance, multi-model comparison, explainability, robustness, external validation and evaluation metrics in the closely related field of chest X-ray pneumonia-detection studies. Symbols: Included, not included, or partial/unclear. Abbreviations: TL = transfer learning, CB = class balancing, MC = multi-model comparison, GC = Grad-CAM, SH = SHAP, RT = robustness testing, EV = external validation.

Study (Year)	Model(s)	TL	CB	MC	GC	SH	RT	EV	Metrics
Kermany et al. (2018) [1]	Inception (TL)	✓	∼	×	×	×	×	×	Acc, Sens, Spec
Rajpurkar et al. (2017) [19]	DenseNet121 (CheXNet)	✓	∼	×	✓	×	×	∼	AUC, F1
Rahman et al. (2020) [27]	VGG, ResNet, DenseNet, SqueezeNet	✓	∼	✓	×	×	×	×	Acc, Sens, Spec
Kundur et al. (2023) [20]	Custom CNN, VGG16, ResNet, Inception, DenseNet	✓	×	✓	×	×	×	×	MAE, Acc
Chamseddine et al. (2022) [17]	CNN + SMOTE/weighted loss	✓	✓	✓	×	×	×	×	Acc, F1, AUC
Jiang et al. (2021) [16]	Improved VGG16	✓	∼	✓	×	×	×	×	Acc, Prec, Rec
Yang et al. (2024) [26]	CNN + background-aware XAI	✓	∼	✓	✓	×	∼	×	Acc, AUC
MDPI (2025) [28]	✓	∼	✓	✓	✓	×	×	×	Acc, interpretability
Frontiers (2026) [29]	VGG16, ResNet50, InceptionV3, Xception	✓	∼	✓	×	×	×	×	Acc, Prec, Rec, F1, AUC
Explainable DL (2025) [24]	DenseNet121 vs ResNet50 + Grad-CAM	✓	∼	✓	✓	×	×	×	Acc
Proposed study	**Custom CNN, VGG16, ResNet50**	✓	✓	✓	✓	×	∼	×	**Acc, Prec, Sens, Spec, F1, AUC, efficiency**

is illustrated in Fig. 13. The process goes from the selection and inspection of the dataset to cleaning, pre-processing, splitting, and augmentation of the data, to the development of a custom CNN, transfer learning, training, evaluation, explainability, comparison with previous models, and interpretation of the results.

3.2 Dataset Description

Subsection Dataset Source and Accessibility The data set for this study is Chest X-Ray Images (Pneumonia) dataset publicly available on Kaggle. This data set was initially published by Kermany et al. [1] and was obtained from the Guangzhou Women and Children's Medical Center. This is available on kaggle at url: https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia. This dataset was chosen because it is a widely used standard for pneumonia detection, is highly annotated by experts and has a sufficient number of images for deep learning.

3.2.1 Dataset Composition

This data set consists of 5,856 JPEG images of a chest X-ray in greyscale. There are two classes (Normal and Pneumonia) and the images are paediatric radiographs of patients between the ages of one and five. The data is available in a set of predefined training, validation and test folders. Expert physicians already removed poor quality and unreadable scans from the dataset, ensuring that this dataset contained high quality scans suitable to the study requirements.The dataset was curated by expert physicians who removed poor quality and unreadable scans ensuring that this dataset contained high quality scans appropriate to the study requirements. Table 1 shows the summary of the splits.

3.2.2 Class Definitions

Normal class refers to chest radiographs with no areas of abnormal opacification. Pneumonia class: There are radiographs demonstrating opacification of the lungs, consolidation or infiltrates typical of bacterial or viral pneumonia. The clinical difference may be slight as the pattern of pneumonia can overlap with other lung conditions and this makes the job difficult.

3.2.3 Dataset Quality Assessment

The first major quality problem is the imbalance in class as Pneumonia images constitute 73% of the data and Normal images 27%. The small validation split given (16 images) gives noisy validation signals during training. Thus the independent test set (624 images) will be considered as the key for the reliable evaluation. Acquisition bias is a potential limitation of generalisation arising from the dataset being from a single medical centre and a paediatric population.

3.2.4 Dataset Visualization

There are representative images from each class shown in Fig. 2. The class-distribution is plotted across all splits, as can be seen in Fig. 1 visually confirms the imbalance mentioned above.

3.3 Data Preparation

3.3.1 Data Cleaning

The dataset was used in its expertly curated form, where images with corruption and unreadability had been filtered by the original authors [1]. No more corrupt or duplicate images were found when loading and all images were successfully read with the data generators.

3.3.2 Data Splitting

The previously defined data set divisions were maintained to ensure a relative comparison with previous studies. The training set contains 5,216 images (1,341 Normal and 3,875 Pneumonia). There are 16 images in the validation set (8 Normal and 8 Pneumonia). There are 624 images (234 Normal, 390 Pneumonia) in the independent test set. The splits are already available at the level of the image and therefore do not require any resplitting of images from the same patient, as does the case with other data sets, to prevent clear data leakage from one split to another.

3.3.3 Image Preprocessing

The images were resized to all be 224×224 pixels to fit the input requirement of the pre-trained backbones. The images were all displayed in 3-channel RGB colour because the images were initially in X-rays channel. The intensities of the pixels were normalised to the range [0,1] by dividing by 255. This normalisation stabilises and speeds up training, as the input value is kept in a small range of constant values.

3.3.4 Data Augmentation

Only the training set was augmented which enhanced generalisation and partially offset the class imbalance. Random rotation of up to $\pm 20°$, width and height shifted by up to 10The nearest-pixel strategy was used for filling in empty regions created by these transformations. These augmentations are suitable for chest X-rays because the small geometric and brightness variations represent a realistic range of patient positioning and image acquisition and do not affect the diagnostic information of the image. Only rescaling was applied to both the validation and test sets, and no augmentation, to be able to fairly assess these sets.

3.3.5 Class-Imbalance Handling

To offset this imbalance, balanced class weights were used in the training process as well as augmentation. The weights were calculated based on the standard balanced heuristic on the training labels. The weight obtained in the Normal class was about 1.945 and in the Pneumonia class was about 0.673. Because the penalty for misclassification of the underrepresented Normal class is higher, than normal class, the models are less likely to classify the underrepresented class as the majority class.

3.4 Model Development

3.4.1 Custom CNN Baseline

The custom CNN is composed of four convolutional blocks of growing depth that take in $224 \times 224 \times 3$ input images. Each of the blocks applies 32, 64, 128 and 256 filters, all with 3×3 kernels, ReLU activation, and same padding. After each convolutional layer, there are batch normalisation and 2×2 max pooling layers which are used to stabilise training and to decrease spatial dimensions. The global average pooling layer combines the feature maps into a feature vector after the convolutional base. This is followed by a fully connected layer with 512 units, the function of which is ReLU, and a regularisation layer in the form of a dropout layer with a dropout rate of 0.5. The last layer is one neuron with sigmoid activation which has an output of probability of Pneumonia class. The architecture is summarised in Table 2 and depicted in Fig. 4.

The model one used for transfer-learning is the VGG16.VGG16 is used for transfer learning. The first model is a transfer-learning model, where the convolutional base is the VGG16 pre-trained on ImageNet [21]. The original top classification layers were detached and the convolutional base was initially frozen allowing the features in the ImageNet to be preserved. A new classification head that includes an average pooling layer across all spatial dimensions, followed by two dense layers (128 and 256 units) using ReLU, dropout (0.3 and 0.5 respectively) and a sigmoid output layer was added. The model was trained first with the base frozen with only the new head being trained. It was then tuned by unfreezing the last few convolution layers of VGG16 base and training it on lower learning rate. The two-stage strategy is designed to make the high-level features adapt to chest X-rays without altering the low-level features.

In the second approach to transfer-learning, ResNet50 was employed.The second approach for transfer learning is using ResNet50. The second instance of transfer learning is to use Resnet50 pre-trained on ImageNet, which adds residual connections that allow for very deep networks [22]. A similar classification head was added to both to make a fair comparison between the two transfer-learning models. The base was also frozen first and unfrozen for head training and then re-frozen for fine-tuning with a lower learning rate, similar to VGG16. By employing the same head with the same training methodology, the sole factor being varied is the backbone architecture itself.

This section describes the process of feature extraction and fine-tuning strategy.In this section, the process of feature extraction and fine-tuning strategy is explained. The same double-stage transfer learning approach was used for both the pre-trained models. The convolutional base was frozen and only trained as a fixed feature extractor in the first stage, while the new head was trained. In the second stage, unfreezing and fine-tuning was performed on the second layer of the base (the layer that is further down), at a lower learning rate so that the representations can be adapted to the medical domain. The latter layers were selected for fine tuning because they represent higher level, task-specific features, while the former layers represent generally visual primitives which transfer without modification.

3.5 Training Procedure

3.5.1 Loss Function

All models were trained with binary cross-entropy loss that is the common loss for the two class classification with a sigmoid output layer. It is appropriate to use cross entropy (CE) here because it directly quantifies the difference between the predicted probability of Pneumonia and the true binary label. To make the objective more sensitive to errors on the minority Normal class, weighting was added to this loss.

Optimizer and Learning Rate All models were trained using the Adam optimiser, which adapts the learning rates for each parameter separately, and has been shown to converge nicely. The initial learning rate of the custom CNN was 1×10^{-3}. The transfer-learning models employed a smaller initial learning rate of 1×10^{-4} for head training, and an even smaller learning rate for fine-tuning to prevent pre-trained weights from being hurt by re-training. A learning-rate reduction approach reduced the learning rate by a factor of 0.5 when the loss on the validation set leveled off.

3.5.2 Training Controls

Throughout, a batch size of 32 was used. The custom CNN was trained for a total of 15 epochs while the transfer-learning models were trained for a total of 10 epochs each. The early stopping was implemented to stop overfitting by giving a patience of 5 epochs while monitoring the validation loss. The best model based on validation accuracy was saved using model checkpointing. To ensure reproducibility, random seeds were set in NumPy and TensorFlow.

3.5.3 Hyperparameter Selection

The values of the hyperparameters were chosen according to standard practice in the literature and optimized within the limit of the Kaggle GPU resources. All the training hyperparameters are provided in Table 3.

3.6 Evaluation Framework

3.6.1 Primary Predictive Metrics

Model accuracy, precision, sensitivity (recall), specificity, F1-score and area under the ROC curve (AUC-ROC) were used to evaluate model performance. In this problem the accuracy is the percentage of all test radiographs that are correctly classified as Normal or Pneumonia as given in Equation 1. Precision is the ratio of images predicted to be Pneumonia and actually being Pneumonia, as in Equation 2. The proportion of true Pneumonia cases that the model correctly classifies, i.e., 100Specificity is the percentage of truly normal cases that are classified as normal (see Equation 4). The F1-score is the harmonic mean of precision and sensitivity (as per Equation 5) and the AUC-ROC represents the overall compromise between sensitivity and false positive rate over thresholds.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$Thesensitivityvaluealsoknownas Recall = TP/(TP + FN) \tag{3}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{4}$$

$$TheF1-scoreisdefinedastheharmonicmeanofprecisionandsensitivity.TheF1scoreistheharmonicaverageofPr \tag{5}$$

In this case, TP refers to the number of true positive, TN to the number of true negatives, FP to the number of false positive and FN to the number of false negative, with Pneumonia defined as the positive class.

3.6.2 Class-Level Evaluation

In addition to the aggregate results, per-class precision, recall and F1-scores were studied for the Normal and Pneumonia class. Confusion matrices were created for each model to reveal the pattern of errors, including the number of pneumonia cases that were missed. Special effort was made on the minority-class (Normal) because it is more difficult to learn because of the imbalance.

3.6.3 Training Behaviour

The accuracy and loss of training and validation were tracked on each epoch. The convergence of these curves was sampled, when early stopping was called, and whether they were overfitting or unstable.

3.6.4 Computational Efficiency

Each model was evaluated for computational properties to evaluate deployment feasibility. The given efficiency metrics are the number of total and trainable parameters, the model size in megabytes and the average inference time per image. These metrics can help in analysing the accuracy-efficiency trade-off explored by RQ5.

3.7 Explainable AI Methodology

3.7.1 Grad-CAM Procedure

The Grad-CAM method was employed to create heatmaps of areas that most affected a prediction on a radiograph, in the form of class-discriminative heatmaps [14]. These weights (gradients) are the gradients of the Pneumonia output with respect to the activations of the final convolutional layer. The map was then normalised and superimposed on the initial X-ray to demonstrate the spatial focus of the model. Confident correct predictions were chosen and incorrect predictions, if applicable, for example cases to check the model's reasoning across situations.

3.7.2 Complementary Attribution Methods

SHAP is a complementary, theory-driven attribution method that gives an attribution score to each region of the image that contributes to predicting the class [15]. It was not calculated in the present experiments and is a suggestion for future research to confirm the Grad-CAM results. A second attribution approach would help to validate the Grad-CAM view of model attention.

3.7.3 Explanation Assessment

Qualitative evaluation of the Grad-CAM explanations was performed by checking whether the highlighted regions are within the lung regions or the background or image artefacts. It is recognized that while visually favorable heatmaps do not constitute correct clinical reasoning, the explanations are considered supporting evidence and not a guarantee. The highlighted areas are therefore used only as markers for the model's focus, and not as a substantiated causal explanation of its choices.

The software, hardware, and reproducibly sections follow.

All experiments have been done in Python with TensorFlow and Keras. Training and evaluation were implemented in the Kaggle notebook environment with GPU acceleration.Training evaluation were performed in Kaggle notebook on GPU acceleration. The key libraries that supported the main libraries were NumPy, pandas, scikit-learn, OpenCV, Matplotlib, and seaborn. NumPy and TensorFlow seeds were set to make the results reproducible. The code is openly available in the project's public GitHub repository while the dataset is the public Kaggle Chest X-Ray (Pneumonia) dataset.

Ethical and Practical Issues

Several ethical and practical considerations are applicable to this project, given that it is related to medical images. The data set is a public de-identified and carefully curated data set, thus reducing direct concerns about privacy. The data is however from one paediatric population, and any model deployed may be biased towards that population, so it may not be generalisable to adults or other settings. Throughout this study, sensitivity is emphasized because the clinical risk of false negatives is great, as a delayed diagnosis of pneumonia may result in delayed treatment. For these reasons, such a system is not meant to be used independently as a diagnostic tool, but is meant to be used in conjunction with the human eye and brain for decision support.

3.8 Dataset

In this study we use only one dataset, which is the Chest X-Ray Images (Pneumonia) dataset described above and summarised in Table 1. Consists of 5,856 paediatric chest radiographs divided into Normal and Pneumonia classes in well-defined training, validation, and test folders, with a significant majority of Pneumonia cases. The class-distribution figure (Fig. 1) together with the representative sample figure (Fig. 2) provide a visual description of the data.

3.9 Detailed Methodology

The overall pipeline combines above and makes one reproducible workflow as a pipeline. The data is first loaded from its built-in splits, checked for class balance, and pre-processed by resizing to 224×224 pixels, transforming to RGB color, and rescaling to $[0, 1]$. Data is then augmented and balanced class weights are used to address class imbalance in the training set. Second, three models are developed and trained using a common protocol: a custom CNN trained from scratch and VGG16 and ResNet50 fine-tuned with a two-step freeze-then-fine-tune transfer-learning process, all with binary cross-entropy loss, the Adam optimiser, early stopping and checkpointing. Third, the trained models are independently tested on the 624-image test set through predictive metrics, confusion matrices, training curves and efficiency metrics and interpreted with Grad-CAM, followed by a comparison of the models to answer the research questions. As can be seen in Fig. 13 the entire workflow is shown.

3.10 Evaluation Metrics

Pneumonia is considered the "positive class" and the evaluation metrics are specific for this pneumonia-detection problem. The measure of accuracy (Equation 1) gives an overall classification rate when the radiograph data is balanced, but is not used alone and may be misleading in the case of class imbalance. Sensitivity (Equation 3) is the proportion of the truly pneumonia-positive radiographs that are correctly identified, which is the key metric as false negatives are the most harmful mistake in this context. Specificity (Equation 4) indicates the percentage of healthy images that are correctly identified and indicates the occurrence of false alarms. Precision (Equation 2) and the F1-score (Equation 5) describe the reliability of positive predictions, and AUC-ROC provides an overview of the discrimination at all decision thresholds.

3.11 Experimental Settings

Wherever possible, experimental conditions were kept the same among the models for a fair comparison. All models were trained on $224 \times 224 \times 3$ inputs, with a batch size of 32, Adam optimiser and balanced class weights loss function. The custom CNN was trained for a maximum of 15 epochs at a learning rate of 1×10^{-3}, with staged fine-tuning of the transfer-learning models at a lower learning rate of 1×10^{-4}. All three methods were used uniformly: early stopping, learning-rate reduction on plateau and model checkpointing. The hyperparameter setting for the entire model is given in Tab. 3 and the custom CNN design is contained in Tab. 2 and Fig. 4.

4 Results

The results are grouped by research questions not in the order of codes. All of the metrics were calculated on the independent test set consisting of 624 images (234 Normal and 390 Pneumonia) and Pneumonia was considered the positive class.

The results of the dataset and pre-processing are presented in the following section.

All 5,856 images loaded were usable and no additional corrupted or duplicate images were found after loading the predefined splits of the dataset. The training set was comprised of 5216 images, the validation set 16 images and the test set 624 images. The training data was highly imbalanced, having 3,875 images of Pneumonia and 1,341 images of Normal. To correct this, weights of 1.945 (Normal) and 0.673 (Pneumonia) were used in the training set, while the latter were only augmented in the training set. Some representative examples of the preprocessed and

Table 1: Description of the Chest X-Ray Images (Pneumonia) dataset used in this study and the distribution of the data set in terms of classes, which reveals a high degree of class imbalance between the Pneumonia and Normal class.

Attribute	**Details**
Total images	5,856 JPEG chest X-rays
Classes	2 (Normal, Pneumonia)
Normal (total)	1,583 (∼27%)
Pneumonia (total)	4,273 (∼73%)
Input resolution	Resized to 224 × 224, RGB
Train split	5,216 (1,341 N / 3,875 P)
Validation split	16 (8 N / 8 P)
Test split	624 (234 N / 390 P)
Source	Guangzhou Women and Children's Med. Center

Table 2: The architecture of the custom CNN baseline, which consists of four convolutional blocks of increasing depth, followed by global average pooling and a dense classification head.

#	**Layer**	**Output Shape**
1	Conv2D(32) + BN + ReLU	$(224, 224, 32)$
2	MaxPool 2 × 2	$(112, 112, 32)$
3	Conv2D(64) + BN + ReLU	$(112, 112, 64)$
4	MaxPool 2 × 2	$(56, 56, 64)$
5	Conv2D(128) + BN + ReLU	$(56, 56, 128)$
6	MaxPool 2 × 2	$(28, 28, 128)$
7	Conv2D(256) + BN + ReLU	$(28, 28, 256)$
8	MaxPool 2 × 2	$(14, 14, 256)$
9	GlobalAveragePooling2D	$(256,)$
10	Dense(512) + ReLU	$(512,)$
11	Dropout(0.5)	$(512,)$
12	Dense(1) + Sigmoid	$(1,)$

Table 3: Hyperparameter configuration of the custom CNN and the transfer-learning models, with optimiser, learning rates, batch size and training controls.

Hyperparameter	**Value**
Input size	$224 \times 224 \times 3$
Batch size	32
Optimizer	Adam
Loss	Binary cross-entropy
LR (Custom CNN)	1×10^{-3}
LR (Transfer learning)	1×10^{-4} (fine-tune lower)
Epochs (Custom CNN)	up to 15
Epochs (Transfer learning)	up to 10 per stage
Class weights	Normal 1.945 / Pneumonia 0.673
Early stopping	patience 5, restore best
LR reduction	factor 0.5 on plateau
Random seed	42

8 8

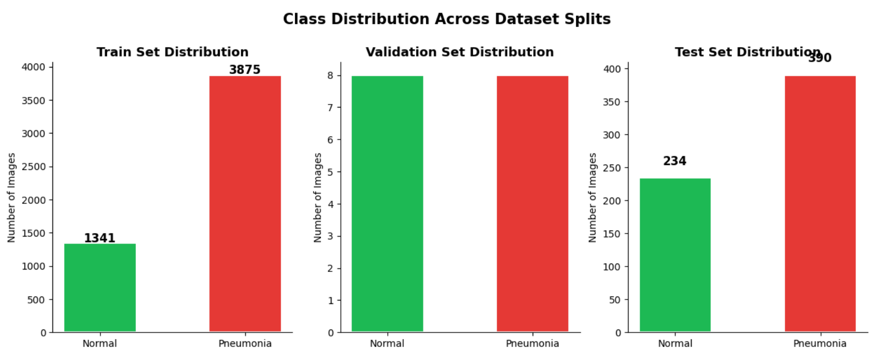

Figure 1: Class distribution across the training, validation and test sets, illustrating that there is a significant imbalance in the numbers of cases and normal in the data set, which is addressed by class weighting and augmentation.

augmented training images are depicted in Fig. 3 and the class distribution over splits is depicted in Fig. 1. These pre-processing results set the imbalanced starting conditions of RQ1 and RQ3.

After training and convergence results

In Figure 5 it can be seen that the accuracy and loss curves for the custom CNN during training and testing are similar.The training and validation accuracy and loss curves for the custom CNN can be seen to be similar in Figure 5. The training accuracy for the custom CNN gradually rose with each epoch and at the end of 15 epochs reached a high training accuracy. The validation signal was noisy and unstable during training, however, which is due to the very small 16-image validation split. The training curve of the transfer-learning models are displayed in Fig. 6 and Fig. 7 respectively, and both achieved high train accuracy in fine-tuning stage. All models had early stopping and learning-rate reduction to prevent overfitting and to stop training when the validation loss failed to decrease.

4.1 Overall Model Performance

Results of the three models on the overall test-set are summarized in Table **??** and the ROC curves in Fig. 8. The VGG16 model as transfer-learning having the highest overall performance with test

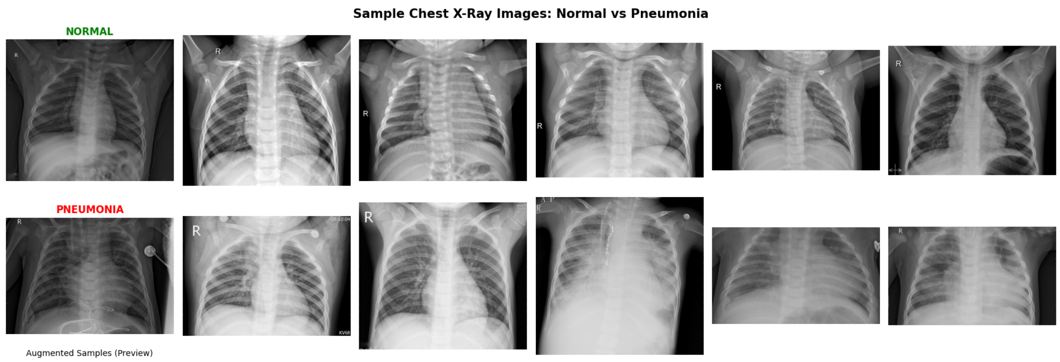

Figure 2: Samples of representative chest X-rays from the two classes, Normal and Pneumonia, to show the visual properties that the models need to learn.

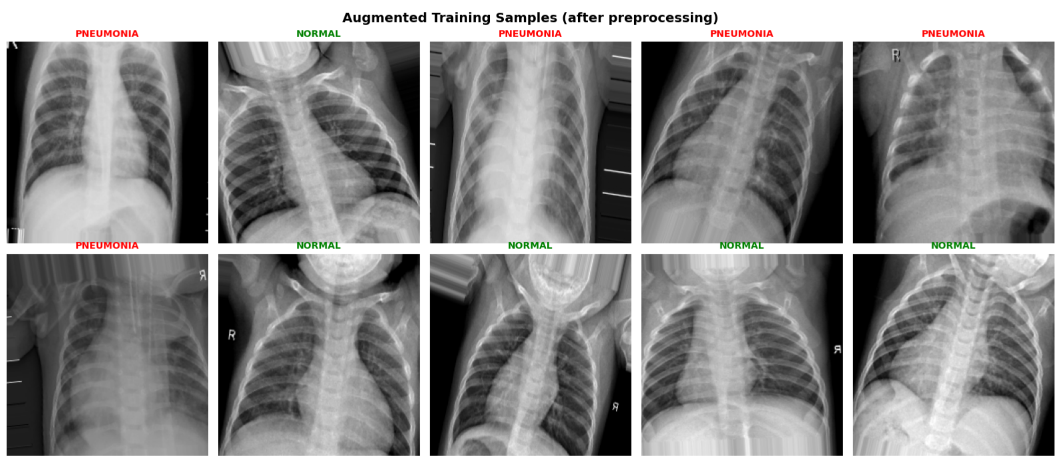

Figure 3: Rotated, translated, magnified, flipped and lighted up/reduced images; examples of augmented training images after pretreatment to enhance the generalisation.

The overall test-set performance of the three models on the 624 image independent test set is reported as accuracy, precision, sensitivity, specificity, F1-score and AUC-ROC, with Pneumonia class as the positive class.

Model	**Acc.**	**Prec.**	**Sens.**	**Spec.**	**F1**	**AUC**
Custom CNN	0.622	0.953	0.415	0.966	0.579	0.882
VGG16	**0.925**	0.905	**0.982**	0.829	**0.942**	**0.974**
ResNet50	0.800	0.932	0.733	0.910	0.821	0.922

accuracy of 0.925, sensitivity of 0.982 and AUC-ROC of 0.974. ResNet50 had the second highest accuracy (0.800) and AUC-ROC (0.922). Although the highest specificity was 0.966 and the highest precision was 0.953, the custom CNN had the lowest accuracy of 0.622 and the lowest sensitivity of 0.415. VGG16 is thus the best model in terms of accuracy, sensitivity, F1-score, and AUC-ROC, as shown numerically. The overall performance results directly answer the question RQ2.

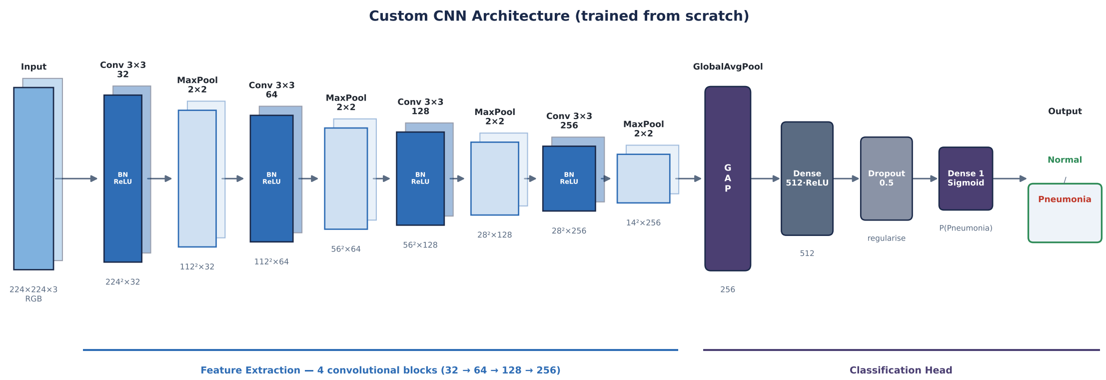

Figure 4: Architecture of the custom CNN baseline, showing the four convolutional blocks (Conv 3×3 + BN + ReLU followed by 2×2 max pooling) of increasing depth (32 → 64 → 128 → 256), global average pooling, and the dense classification head ending in a single sigmoid output.

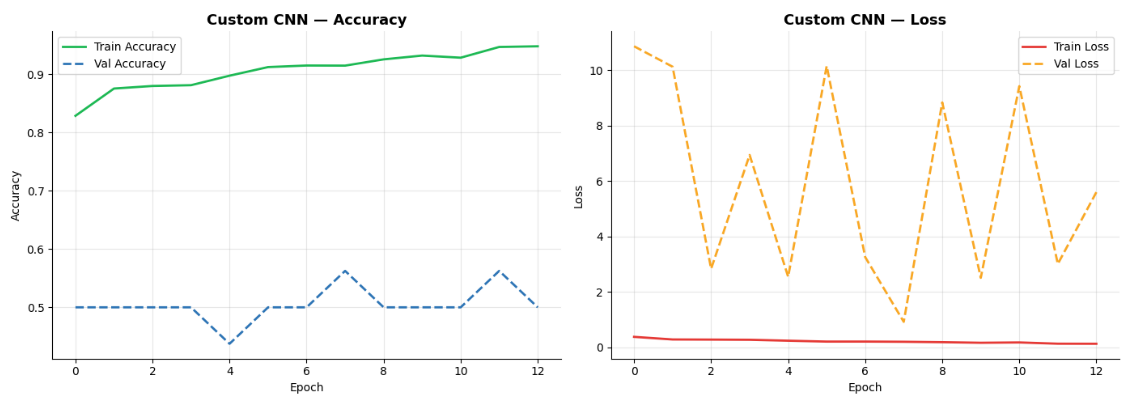

Figure 5: Training and validation accuracy and loss curves for the custom CNN, with steadily increasing training accuracy and noisy validation signals due to the extremely small size of the 16 image validation split

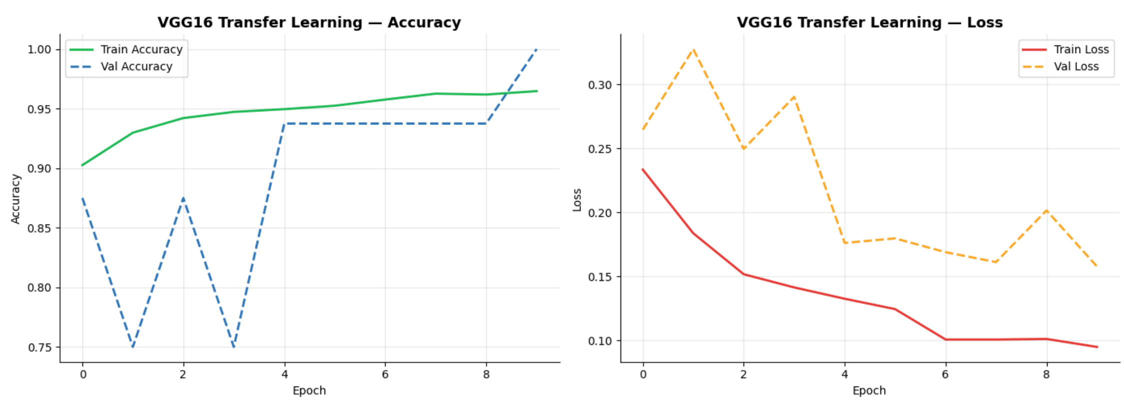

Figure 6: Training and validation curves of the VGG16 transfer-learning model in the frozen-base and fine tuning phases.

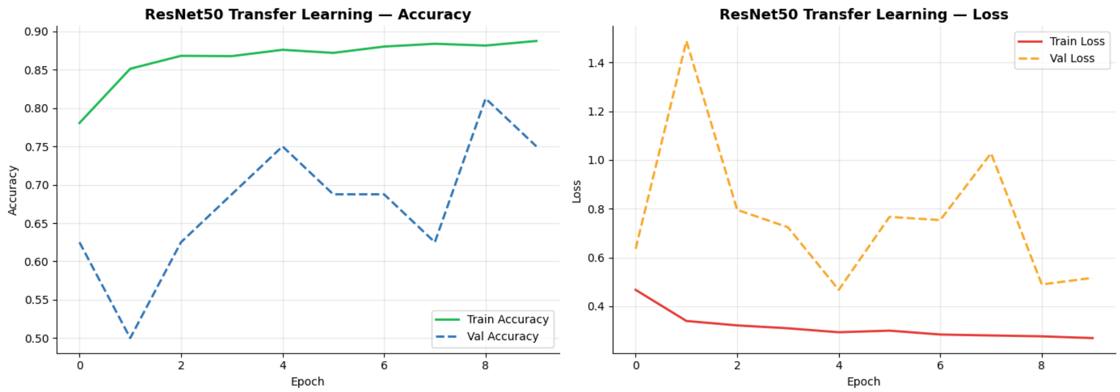

Figure 7: Frozen-base and fine-tuning training and validation curves for ResNet50 transfer-learning model.

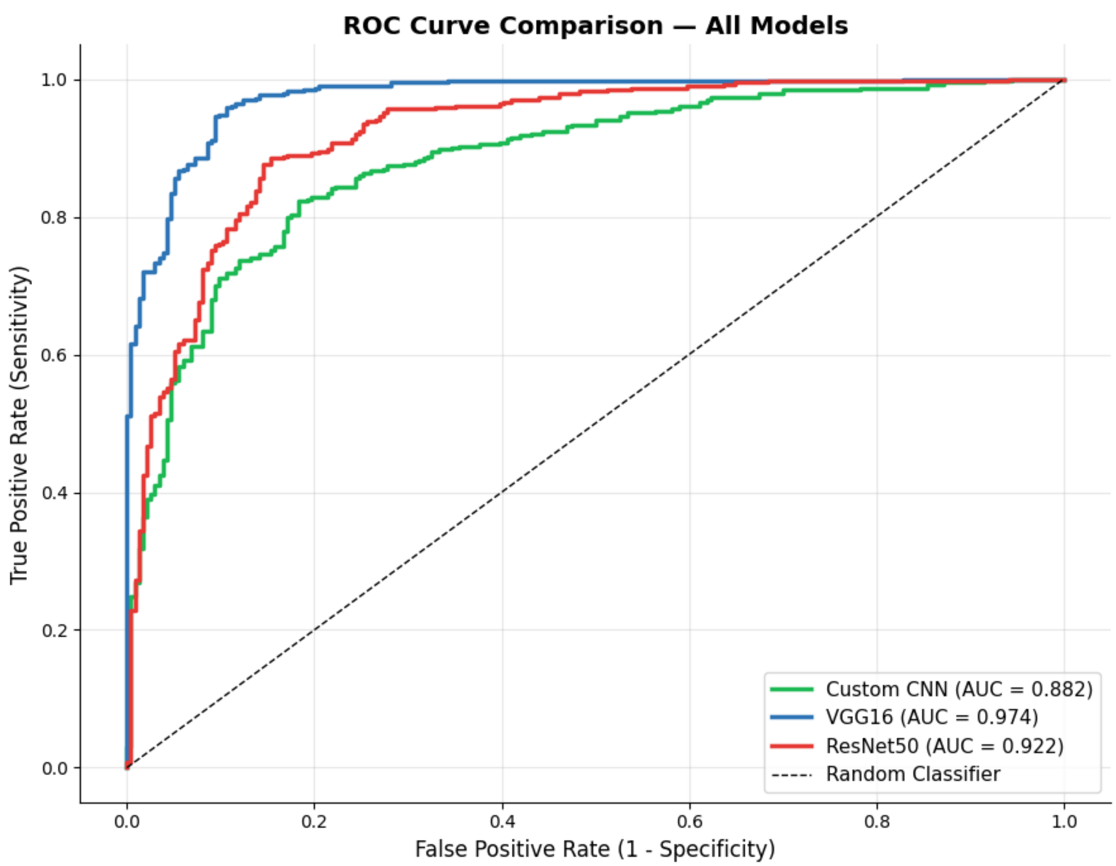

Figure 8: ROC curves of the three models in the independent test set. The custom CNN with an AUC value of 0.882 is followed by ResNet50 with an AUC value of 0.922 and VGG16 with the highest AUC value of 0.974.

4.2 Class-Level Performance

The confusion matrices of the three models are illustrated in Fig. 9, Fig. 10 and Fig. 11. For the custom CNN, there was 162 correctly detected Pneumonia cases out of 390 Pneumonia cases, 228 false-negatives (missed pneumonia cases), 226 correctly classified Normal cases out of 234 Normal cases. VGG16: 383 out of 390 Pneumonia cases were correctly predicted and 234 out of 240 Normal

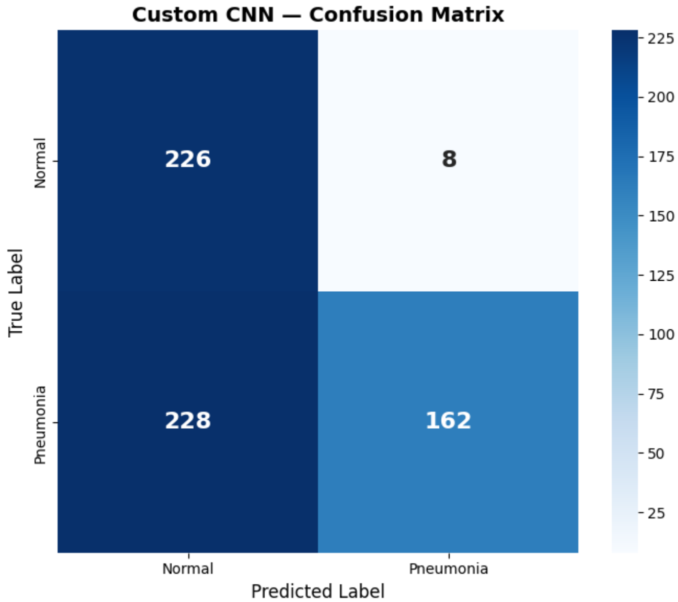

Figure 9: Confusion matrix for the custom CNN on the test set of 390 pneumonia cases, where the CNN missed 228 of these cases while correctly classifying the majority of Normal cases.

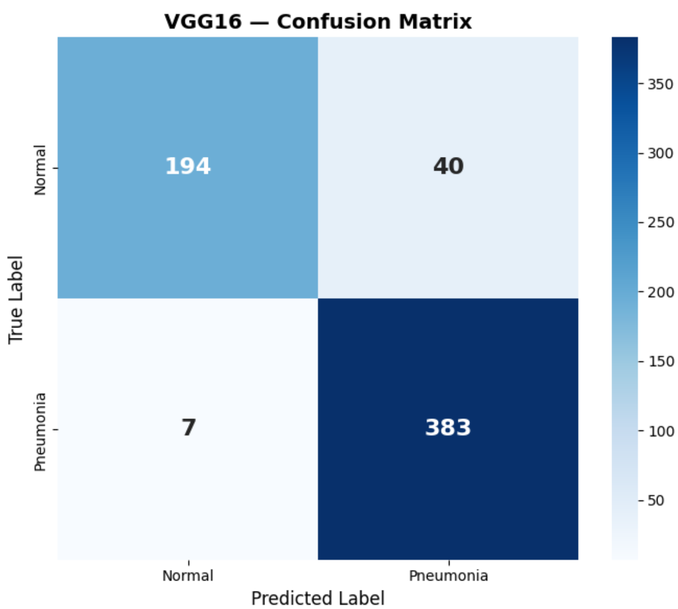

Figure 10: Confusion matrix of the best model (VGG16) with 383/390 correctly identified pneumonia cases and 7 false negatives.

cases were correctly predicted (40 false positives). In ResNet50, 286 out of 390 Pneumonia cases were correctly classified and 104 out of 234 Normal cases were misclassified.In ResNet50, 286 out of 390 cases of Pneumonia were correctly detected and 104 out of 234 cases of Normal were not. On a class level, the Pneumonia (minority-error) behaviour clearly favors the VGG16 model as it has the lowest number of pneumonia cases it fails to detect. The class-level results pertain to RQ3 considering the performance under the condition of class imbalance.

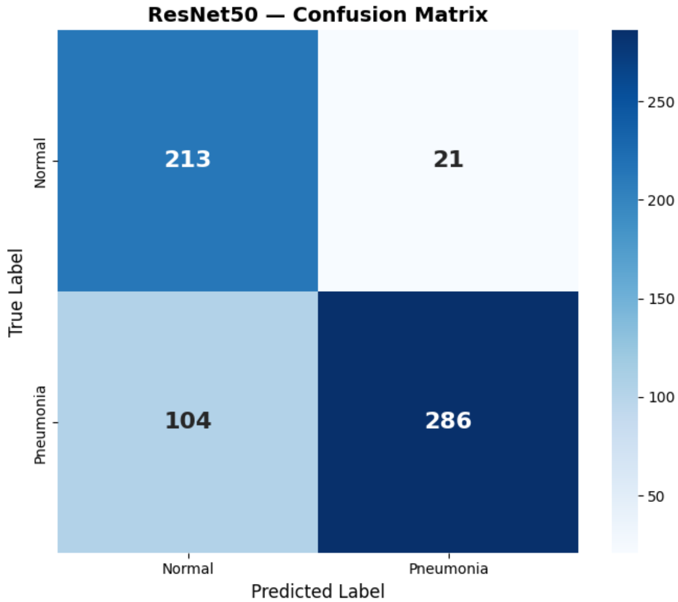

Figure 11: Confusion matrix of ResNet50 model with 390 test cases, where 104 cases were not correctly diagnosed.

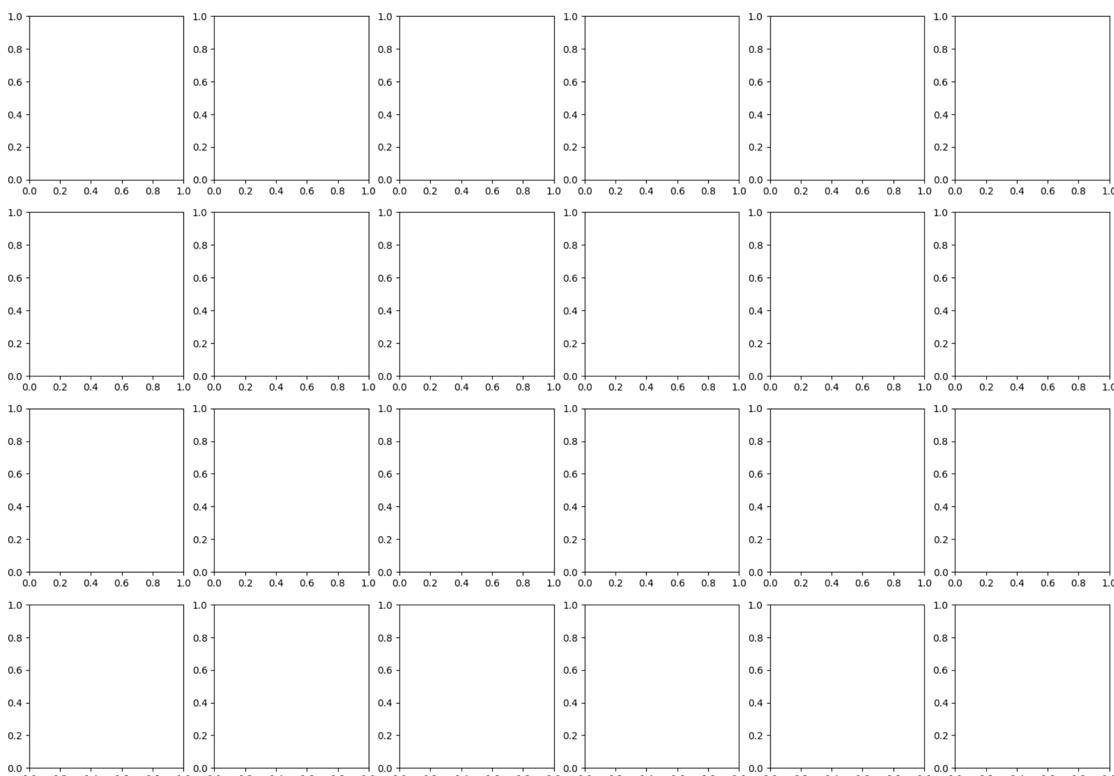

Figure 12: Heat maps overlaid on chest X-rays for the selected examples, showing the most important areas of the lungs used for the prediction of Pneumonia.

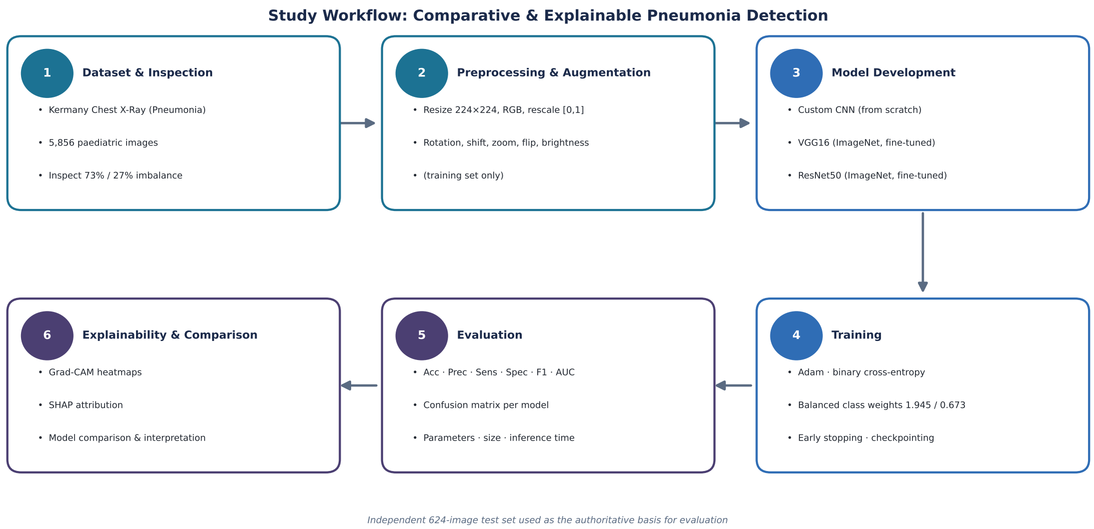

Figure 13: The overall workflow of the study, including dataset preparation and inspection, the preprocessing and augmentation of the dataset, development of the model (custom CNN, VGG16, ResNet50), training, evaluation, and explainability of the model with Grad-CAM and SHAP.

4.3 Explainable AI Results

4.3.1 Grad-CAM Results

Representative Grad-CAM heatmaps are shown in Fig. 12. The heatmaps indicate the regions of the lungs which had the greatest influence on the Pneumonia predictions for the example radiographs selected. In cases of pneumonia, the highlighted areas tend to be within the lung fields and not the background of the image (as shown here).

4.3.2 Complementary Explanation Methods

In this work, we primarily centered our attention on Grad-CAM as the explainability technique for visualizing model attention. A complementary method is SHAP-based attribution (SAA), which computes positive and negative contributions to different regions of an image that was not performed in the current experiments. It is identified as an area for future research to compare the results of the Grad-CAM method with a second method based on theory.

4.3.3 Explanation Assessment

Grad-CAM heatmaps were evaluated by examining the highlighted area compared to the background or image artefacts by looking if the area is inside the lung field. The attention was mainly focused in the regions of the lung for correctly classified cases of pneumonia, which indicates the clinical relevance of the model's predictions. The explainability results cover RQ4 and explainability in RQ5.

4.4 Efficient Deployment and Results

The efficiency of the three models is reported in Table **??** Among the three models, the custom CNN is definitely the lightest, as it has only 522,433 parameters, a model size of 1.99 MB, and

Caption: Computational efficiency of the three models, including total number of parameters, trainable number of parameters, model size in megabytes, and mean inference time per image (mean ± std dev over 50 runs).

Model	Total Params	Trainable	Size (MB)	Inference (ms)
Custom CNN	522,433	521,473	1.99	80.55 ± 3.41
VGG16	14,879,041	7,243,777	56.76	86.62 ± 5.59
ResNet50	24,145,281	5,023,233	92.11	89.53 ± 3.89

an average of 80.55 ms per image for inference time. VGG16 has 14,879,041 total parameters (7,243,777 trainable), a model size of 56.76 MB, and an average inference time of 86.62 ms per image. ResNet50 is the biggest model in which there are 24,145,281 total parameters, model size of 92.11 MB, and the average inference time per image is 89.53 ms. Thus the custom CNN (smallest and fastest) is the least accurate model and VGG16 (intermediate) is the most accurate. These efficiency results, combined with the performance results in Table **??**, directly address the trade-off in RQ5.

A comparison of the present study with published studies is provided here.

VGG16 was the best model in this study, with a test accuracy of 0.925 and test AUC-ROC of 0.974 on the Kermany chest X-ray test set. This is on the same order of magnitude as the range of transfer learning results found in the literature for the same dataset [7]. However, there is limited direct numerical comparison with other studies due to uncertainties in the data splitting, preprocessing and evaluation methods reported. A comparison of the findings to specific published studies is included with the literature matrix in Section 2.

The findings from the research questions are summarized below.

The evidence and main finding supporting each research question is summarized briefly in Table **??**.

5 Discussion

The Discussion is the section of the document that gives a meaning to the results found in Section 4, relates them to the research questions, and evaluates their practical value. It also takes into account the advantages, drawbacks and qualifications that restrict the conclusions.

To interpret the given dataset and draw conclusions from the results of the preprocessing.

RQ1 and RQ3 focussed on the preprocessing strategy required to get usable results from an imbalanced data. Data augmentation and class weight balancing were to avoid the models from just predicting the majority class (Pneumonia). The criterion of high sensitivity of the best model VGG16 indicates that this approach was able to create a learnable minority class (Normal) for pneumonia detection. In parallel, the poor performance of the custom CNN indicates that augmentation and class weighting alone is not sufficient to ensure good minority class behaviour, as the model architecture is also important. The small size of the validation split (only 16 images) was indeed a limitation of the dataset that led to a noisy validation curve and a hard training phase model selection. In this context it was considered that the independent test set of 624 images, which is as it is generally used in literature [7], is the "standard" around which the set of evaluations was built. In general, the results of preprocessing the data show that the data set can be used for this task, but it is an imbalance and a small validation set that are real limitations of the study.

In this section, the performance of CNN models is compared.

The comparison of the three models is the direct answer to RQ2, and is the central result of this study. The results of transferring the CNN to the new dataset were clearly better than the custom CNN, with the VGG16 having an accuracy of 0.925 compared to 0.622 and a sensitivity of 0.982 compared to 0.415. This substantial gain is in line with the previously reported success of ImageNet pretrained features when applied to medical imaging despite the differences between the appearance of chest X-ray and natural images [9, 7]. An interesting finding is that although ResNet50 is deeper than VGG16, the latter performed better on this problem. One possible reason is that the shallower VGG16 was more readily able to fine-tune the limited and imbalanced data, while the deeper ResNet50 required more attention in order to effectively fine-tune the data within the allotted training budget. This is the idea that don't assume that bigger is always better, especially with smaller amounts of data. Finally, the importance of not assuming the highest accuracy is the best model alone is highlighted, as sensitivity and the clinical cost of errors are emphasised in this study. However, when these different models were compared on all clinically important criteria, VGG16 outperformed all the others, and was the clear choice for a recommendation.

Class-Level Reliability and Error Patterns

To answer RQ3, the class-level analysis was conducted to explain the low performance of the custom CNN while having a high accuracy score, although it is not explicitly so classified. The Sensitivity of the custom CNN with respect to pneumonia was only 41.5This pattern suggests that the custom CNN started to overestimate the class Normal and underestimate the majority of pneumonia cases. This is the most dangerous failure mode in a screening situation as each time an individual has pneumonia and is not identified by a screening test, a patient is incorrectly reassured. In comparison, VGG16 had 40 false positives, and only missed 7 of 390 pneumonia cases – a much safer combination for a screening tool. This high precision, therefore, is deceptive in isolation and shows the need to consider accuracy and precision in addition to sensitivity in imbalanced medical tasks in the case of the custom CNNs. The clinical usefulness of a model is strongly related not only with the number of mistakes made but also with the kind of mistakes made, as these patterns of error illustrate.

5.1 Interpretation of Grad-CAM

The explainability analysis is related to RQ4 and to the interpretability part of RQ5. Overall, the Grad-CAM heatmaps tended to focus on the lung regions for correctly classified pneumonia cases, as is expected and desired. This offers qualitative confirmation that the predictions of the best model are not affected by image artefacts but rather by clinically relevant areas. However, it is important to remember that a convincing heatmap does not necessarily mean that clinical reasoning is accurate, as there are many methods and tools used for heatmap generation and these can be sensitive to preprocessing and unstable. The explanations are thus considered evidence of the location of the model, and not a true causal account of the model's decisions. To support this evidence, a complementary method of attribution, like SHAP, should be cross checked with the findings of the Grad-CAM is recommended for future work. This difference is significant for a domain as sensitive as medical imaging and any such system deployed would have to be validated by domain experts to explain its explanations.

5.2 Efficiency and Realistic implementation

The efficiency analysis directly tackles the trade-off in RQ5. The custom CNN is very light weight (only 1.99 MB) and the inference speed is the fastest one, however, the low sensitivity makes it unable to be used for clinical screening even if it is efficient. VGG16 is the medium-size model that achieved the highest accuracy of 97.25In this case ResNet50 is the largest and slowest model (92.11 MB) and did not achieve the highest accuracy, so it is the least appealing efficiency/lower performance trade off in this case. The main practical take-away is that the smallest model is not the best, and the largest model is not the best—and that VGG16 offers the best accuracy/Resources ratio. If deployed in the web or cloud, VGG16 size and speed will not pose a limiting constraint. In future work, an architecture with less resources may be considered for the edge or mobile devices, with the caveat that its sensitivity can be reduced to an acceptable level.

A comparison of what is found in the literature is given as a comparison.

The results for the best performing model in this study closely match the previously reported results of previous transfer-learning experiments on the same chest X-ray data set [7]. The current study is distinguished from numerous previous studies as it integrates a fair comparison across models, explicit class-imbalance treatment, explainability insights from Grad-CAM and an efficiency study in a reproducible pipeline. It is difficult to directly compare with other studies since reported results depend on the splits of the data sets, the preprocessing used, and the evaluation protocols. Therefore, no argument for superiority is advanced over previous work and the contribution is expressed as a careful, transparent and explainable comparison.

The practical and the scientific contributions.

The contribution of the study can be divided into scientific and practical. **Scientific contributions.** The study systematically and like-for-like compares a custom CNN with two Transfer-learning models, both under the same imbalance-aware conditions. It provides class-level reliability analysis with a focus on sensitivity, and Grad-CAM explainability analysis to assess if the models focus on clinically relevant regions. **Practical contributions.** The study illustrates a trainable pipeline that can be used to prioritize and prioritize pneumonia cases for expert review. It quantifies the accuracy-efficiency trade-off, as well, allowing for specific recommendations as to which is the best model for real-world decision support.

Ethical, Social and Safety Implications

The results give rise to a number of ethical and safety issues. The data used is from a single paediatric population, therefore the information used to form a model could be biased and may not be applicable to adults, or to images taken at different equipment and settings. The single worst outcome is a false negative, as this might result in reassurance and a delay in treatment for people who actually have pneumonia. That is why sensitivity is emphasised in the study and why the weak custom CNN is deemed not to be appropriate. If such a system were deployed in practice, it would need to be overseen by competent humans and it would be used as a decision support system, not as a decision maker. Over-reliance on an automated prediction, without clinical confirmation, would be inappropriate for a high-risk medical task of this kind.

5.3 Limitations

Several limitations were noted in this study. This dataset includes only paediatric patients aged 1-5 years from one medical centre, making the results less generalizable. The given validation split has a small number of images (16), resulting in noisy validation signals and less in-training model selection. No external testing on independent data set, only one train/validation/test split, and only one public data set was used. The configuration of the custom CNN was not exhaustively optimised, so this low sensitivity could be due to suboptimal training of the network, but not necessarily due to the limits small custom networks. Kaggle is limited to computational resources within the Kaggle environment, meaning the number of hyperparameters that could be varied or additional experimentation run was limited. The clinical validity of the explanations has not been reviewed by a domain expert in terms of the model's predictions and explanations. Such restrictions do not affect the basic comparative conclusion, but they do narrow the extent of the conclusions.

5.4 Future Research Directions

The following are several realistic extensions to this work. The strong performance of VGG16 on this dataset would be further tested by external validation with independent chest X-ray datasets, to determine if the performance extends to other sources. Other backbones can be tested including DenseNet and EfficientNet to explore if they can deliver better accuracy-efficiency compromise. The CNN could be retrained with tuning the threshold and doing a detailed hyperparameter search to see if the sensitivity of the CNN can be increased. SHAP-based attribution could be used along with Grad-CAM and cross-checked the regions that contribute to predictions. The extension of the task from a binary classification to the distinction between bacterial and viral pneumonia would add to the usefulness in the clinic. Last but not least, the explainability validation would be further strengthened by more rigorous explainability validation including quantitative explainability metrics and domain experts assessment before deployment.

6 Conclusion

This paper proposed a comparative and explainable CNN-based framework to automatically classify paediatric chest X-ray (CXR) images as Normal or Pneumonia. A custom CNN trained from scratch, a VGG16 transfer-learning model and a ResNet50 transfer-learning model were designed and tested in the same imbalanced environment. It used a two-phase freeze-then-fine-tune approach and balanced class weighting for the transfer-learning models, along with image preprocessing and augmentation on the 1,624-image test set, which was kept separate. The model whose performance was closest to the best and most clinically relevant was the VGG16 model, with a test accuracy of 0.925, a test sensitivity of 0.982 and AUC-ROC of 0.974, and a failure rate of only 7 cases of pneumonia out of 390. The small custom CNN, although trained from scratch, was still not enough for such a challenging task as this, with the accuracy of 0.622 and the sensitivity of only 0.415. Overall, the principal conclusion is that transfer learning, and in particular using VGG16, offers a significantly better and safer support for pneumonia screening than a custom CNN, and that, in this context, sensitivity is the most important criteria for selection. For correctly classified instances, the best model's attention was mostly concentrated in the lung fields using the Grad-CAM analysis, which was a qualitative indication that its predictions were based upon clinically meaningful regions. When considering direct clinical use, the VGG16 model provides the best predictive performance and computational efficiency of all the models tested and is a good basis to create a pneumonia screen support tool where humans will be involved. The main limitation of this study is that it uses a single paediatric dataset with a small validation split and no external validation, limiting the generalisability of the results. External validation on independent chest X-ray datasets, as well as the exploration of additional backbones and more rigorous, expert-assessed explainability is valuable direction for future work.

Declarations

The author employed the generative AI tool Claude (Anthropic, San Francisco, CA, USA) to help refine the language, structure and clarity of this manuscript. The author did all the reviewing and editing of the manuscript, and takes the full responsibility for the final version.

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Summary of the results from the research questions listed with evidence used and main result for each question according to the test-set results measured.

Describe the research question(s) and their evidence, and the main results. RQ1	Custom CNN test accuracy and sensitivity (Table ??)	No; the custom CNN obtained only 0.622 accuracy which is less than the 90% target. Table ?? and Fig. 8 show the comparison between RQ2
Three-model and Transfer learning, which was superior with VGG16 showing the highest accuracy: 0.925 and AUC: 0.974. High sensitivity for VGG16 was achieved by using RQ3	Class weighting + augmentation, class level results (confusion matrices)	Balancing enabled, and the custom CNN under-detected pneumonia. For correct cases, the attention of RQ4
Grad-CAM heatmaps (Fig. 12) fall in the lung areas, while the attention of the best model	Performance vs. efficiency (Tables ??, ??)	Smallest model is not the best, VGG16 is the best performance in the moderate size.